AKSHAR AI

SUMMER INTERNSHIP REPORT

WEEK 1 — 25 MAY 2026 – 29 MAY 2026

Project Title:

Chest X-Ray Pneumonia Detection and Analysis System Using Deep Learning

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1. Introduction

Pneumonia is a respiratory disease that causes inflammation in the lungs and can lead to symptoms such as cough, fever, chest pain, and difficulty breathing. Early diagnosis is important to prevent severe complications and improve patient outcomes.

Chest X-ray imaging is one of the most commonly used methods for diagnosing pneumonia. It allows doctors to examine the condition of the lungs and identify abnormalities such as opacities or infected regions that may indicate the presence of pneumonia.

However, manually analyzing large numbers of chest X-ray images can be time-consuming and challenging. The accuracy of diagnosis depends on the experience of radiologists, creating a need for intelligent systems that can assist in medical image analysis.

Recent advancements in Artificial Intelligence (AI) and Deep Learning, particularly Convolutional Neural Networks (CNNs), have shown promising results in medical image classification and disease detection. These technologies can automatically identify important patterns within medical images and support faster analysis.

As part of the internship at Akshar AI Ltd., a Chest X-Ray Pneumonia Detection and Analysis project was initiated to explore the use of AI in healthcare. During Week 1, the primary focus was on dataset acquisition, verification, metadata extraction, exploratory data analysis (EDA), and the development of an initial React-based dashboard for presenting analytical results.

2. Day-wise Work Summary

25 May 2026 (Monday)

- Understood the project objectives and internship requirements.
- Acquired the Chest X-Ray Pneumonia dataset from Kaggle.
- Set up the development environment and organized the project structure.

26 May 2026 (Tuesday)

- Performed dataset verification and validation.
- Checked dataset structure, image counts, and class distribution.
- Conducted integrity checks to identify corrupted or missing files.

27 May 2026 (Wednesday)

- Extracted image metadata including dimensions and file properties.
- Visualized sample chest X-ray images from NORMAL and PNEUMONIA classes.
- Studied dataset characteristics and image variability.

28 May 2026 (Thursday)

- Performed Exploratory Data Analysis (EDA) on the dataset.
- Generated class distribution graphs and resolution analysis plots.
- Conducted pixel intensity analysis to understand image characteristics.

29 May 2026 (Friday)

- Summarized analytical findings and observations.
- Developed an initial React-based dashboard prototype.
- Integrated dataset statistics and visualizations into the dashboard interface.

3. Methodology

The methodology followed during Week 1 focused on understanding, validating, and analyzing the Chest X-Ray dataset before proceeding to machine learning model development. The workflow was divided into five phases: dataset acquisition, dataset verification, metadata extraction, exploratory data analysis, and dashboard development.

Phase 1: Dataset Acquisition

The Chest X-Ray dataset was obtained from Kaggle and organized into three primary partitions: Training, Testing, and Validation. Each partition contained two classes, namely NORMAL and PNEUMONIA. The dataset structure was reviewed to ensure that all required folders and image files were available for analysis.

Phase 2: Dataset Verification and Validation

Python-based validation scripts were developed to verify the integrity of the dataset. File counting operations were performed to determine the number of images present in each partition and class. Additional validation checks were conducted to identify corrupted, missing, or zero-byte image files. This phase ensured that the dataset was complete and suitable for further processing.

Phase 3: Metadata Extraction

A representative set of chest X-ray images was analyzed to extract important metadata such as image dimensions, file format, and channel information. This process provided insights into image variability and helped identify the need for image standardization during future preprocessing stages.

Phase 4: Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the statistical and visual characteristics of the dataset. Sample image visualizations were generated to examine differences between NORMAL and PNEUMONIA classes. Class distribution analysis was conducted to identify dataset imbalance, while resolution analysis and pixel intensity analysis were performed to study image properties and data quality.

Phase 5: Dashboard Development

The analytical outputs generated during the EDA phase were integrated into a React-based dashboard prototype. The dashboard was designed to present dataset statistics, visualizations, and analysis results in a structured and user-friendly format. This phase established the foundation for a future web-based pneumonia detection and analysis system.

4. Tools and Technologies Used

Category	Tool / Technology	Purpose
Programming Language	Python 3.x	Data loading, dataset validation, metadata extraction, and exploratory data analysis.
Data Analysis	Pandas, NumPy	Data manipulation, statistical analysis, and numerical computations.
Image Processing	PIL (Pillow), OpenCV	Loading images, extracting image metadata, and performing image-related operations.
File Management	os, pathlib, glob	Directory traversal, file counting, and dataset verification.
Visualization	Matplotlib	Generation of sample image grids, histograms, and resolution analysis plots.
Visualization	Seaborn	Creation of class distribution charts and statistical visualizations.
Frontend Development	React.js	Development of the dashboard interface for displaying analytical results.
Backend Development	Flask	Backend framework for future API integration and data processing.
Data Source	Kaggle Dataset	Source of the Chest X-Ray Pneumonia dataset used for analysis.
Version Control	Git and GitHub	Source code management, project backup, and collaboration.
Development Environment	Visual Studio Code	Code development, debugging, and project management.
Notebook Environment	Jupyter Notebook / Google Colab	Execution of analysis scripts and visualization of outputs.

Table 1: Tools and Technologies Used

5. Dataset Analysis

The Chest X-Ray dataset was analyzed to understand its structure, quality, and statistical characteristics before implementing machine learning models. This analysis phase focused on validating the dataset, examining image properties, identifying potential issues, and generating visual insights that will support future model development.

5.1 Dataset Verification

The dataset was organized into three standard partitions: Training, Testing, and Validation. Each partition contained two classes: NORMAL and PNEUMONIA.

The verification process confirmed that all required folders were present and accessible. The image distribution across the dataset was as follows:

Dataset Split	Class	Number of Images
Train	Normal	1341
Train	Pneumonia	3875
Test	Normal	234
Test	Pneumonia	390
Validation	Normal	8
Validation	Pneumonia	8

Table 2: Dataset Structure Verification

The verification process confirmed that the dataset was correctly organized and suitable for further analysis.

5.2 Dataset Integrity Check

A dataset integrity validation process was performed to identify corrupted files and zero-byte images that could negatively affect future analysis and model training.

The validation process produced the following results:

- Corrupted Images: 0
- Zero-byte Images: 0

The absence of corrupted or empty image files indicates that the dataset is clean and reliable for further processing. This ensures that future preprocessing and machine learning tasks can be performed without interruptions caused by invalid image files.

5.3 Metadata Analysis

Metadata extraction was performed on selected Chest X-Ray images to understand the structural properties of the dataset. The extracted metadata included image class labels, filenames, image dimensions, channel information, and file formats.

The analysis revealed that the dataset contains images with varying resolutions. Sample images ranged from dimensions such as 816 × 584 pixels to 1846 × 1676 pixels. All examined images were stored in JPEG format and contained three color channels.

The variation in image dimensions indicates that the dataset was collected from multiple imaging sources and equipment configurations. Such differences may introduce inconsistencies during model training if not properly addressed. Therefore, image standardization and resizing will be necessary before implementing machine learning and deep learning models.

The metadata extraction process also confirmed that all images were readable and contained valid information required for further analysis and preprocessing.

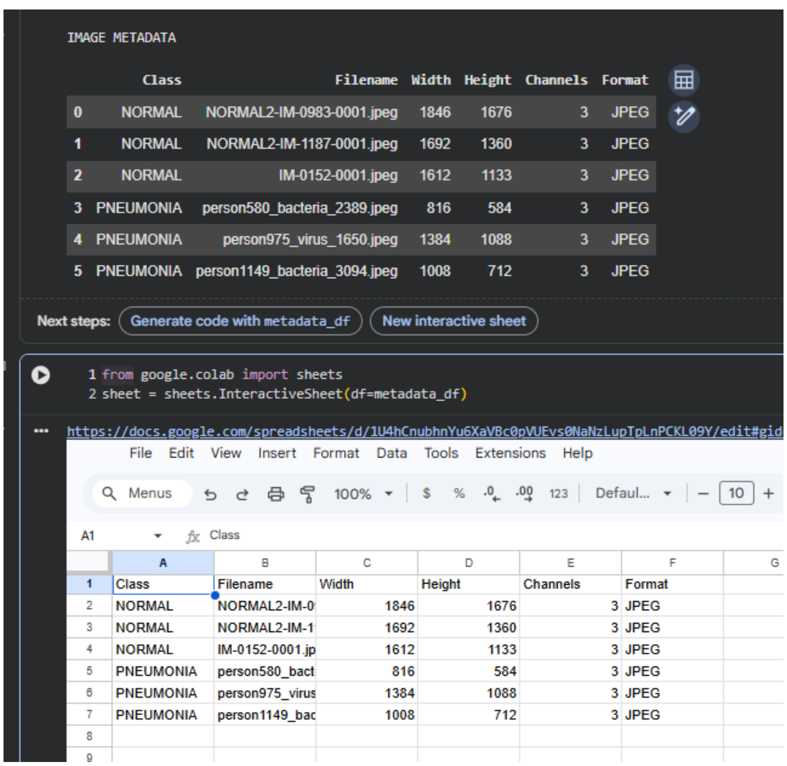

Figure 1: Metadata Analysis Results

5.4 Sample Image Visualization

Sample images from both the NORMAL and PNEUMONIA classes were visualized to gain an initial understanding of the dataset characteristics.

The visualization process revealed noticeable differences between healthy and infected chest X-rays. Images belonging to the NORMAL class generally displayed clearer lung fields with fewer visible opacities. In contrast, images categorized as PNEUMONIA often exhibited denser regions, increased opacity, and abnormal patterns within the lung area.

Visual inspection is an important preliminary step in medical image analysis because it helps verify the correctness of dataset labels and provides insight into the features that machine learning models may later learn to identify.

The sample visualization also confirmed that the dataset contains sufficient visual diversity, which is beneficial for developing robust predictive models.

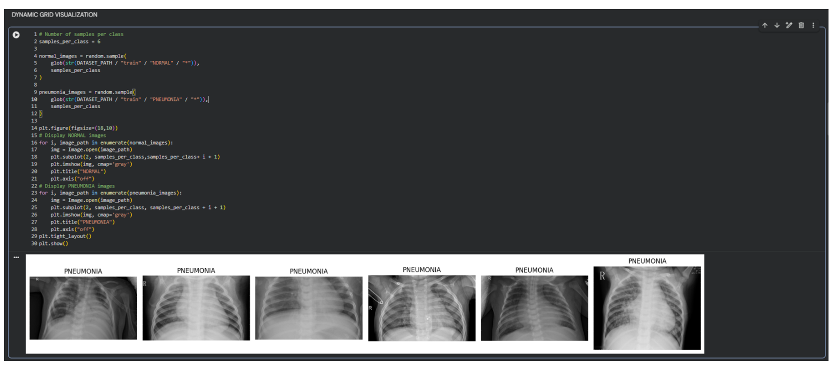

Figure 2: Sample Chest X-Ray Images from NORMAL and PNEUMONIA Classes

5.5 Class Distribution Analysis

Class distribution analysis was performed to examine the balance between NORMAL and PNEUMONIA images within the training dataset.

The analysis showed that the training dataset contains 1,341 NORMAL images and 3,875 PNEUMONIA images. This results in an imbalance coefficient of approximately 2.89, indicating that pneumonia cases significantly outnumber normal cases.

The class distribution graph clearly demonstrates the unequal representation of the two categories. Such class imbalance can influence model performance by causing the learning algorithm to favor the majority class during training.

This observation highlights the importance of applying suitable preprocessing strategies in future stages of the project, such as data augmentation, class weighting, or balanced sampling techniques, to ensure fair and reliable model performance.

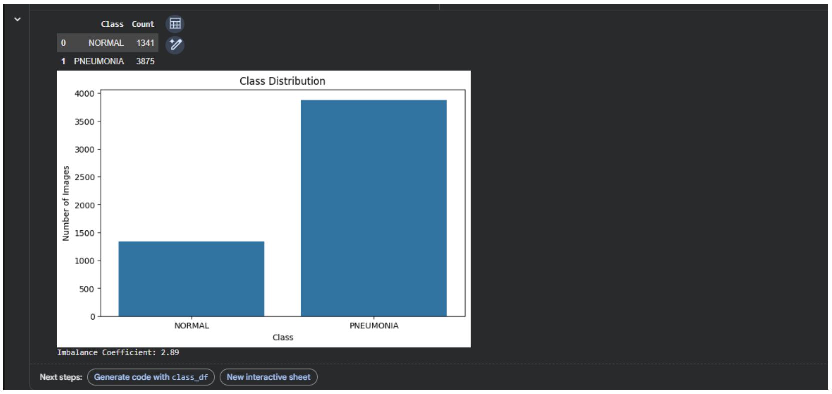

Figure 3: Class Distribution of Chest X-Ray Dataset

5.6 Resolution Analysis

A resolution trajectory analysis was conducted to investigate the relationship between image width and height across both classes.

The scatter plot revealed substantial variation in image dimensions throughout the dataset. Images belonging to both NORMAL and PNEUMONIA classes were distributed across a wide range of resolutions, indicating that no fixed image size was used during data collection.

The presence of heterogeneous image dimensions reinforces the need for a standardized resizing process before model development. Standardizing image dimensions helps ensure consistency in feature extraction and improves computational efficiency during training.

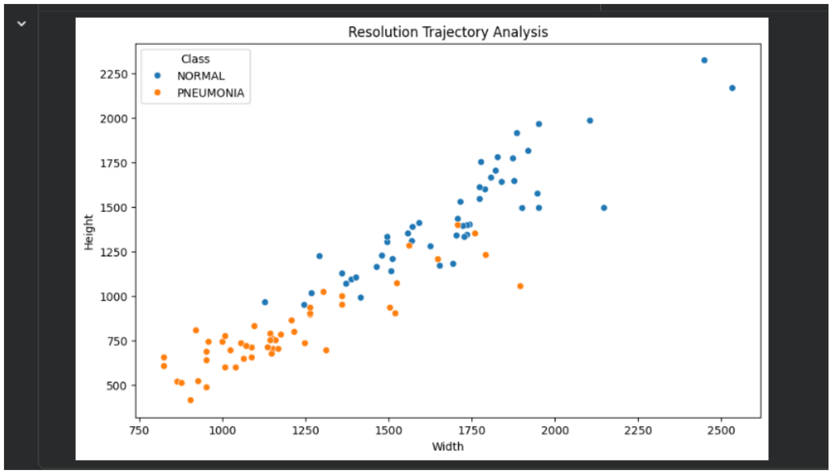

Figure 4: Resolution Trajectory Analysis

5.7 Pixel Intensity Analysis

Pixel intensity distribution analysis was performed to examine grayscale value patterns present in the chest X-ray images.

The histogram comparison between NORMAL and PNEUMONIA classes revealed differences in pixel value distributions. While both classes exhibited overlapping intensity ranges, pneumonia images showed variations in density patterns that may correspond to lung abnormalities and opacities.

The observed differences suggest that pixel intensity information contains meaningful diagnostic features that can be utilized by machine learning and deep learning models for classification tasks.

This analysis provides an initial indication that image-based features may be effective for distinguishing between healthy and infected lungs, thereby supporting the feasibility of future automated pneumonia detection systems.

Characteristic	NORMAL Images	PNEUMONIA Images
Pixel Distribution	More concentrated in lower to mid intensity ranges	More widely distributed across intensity ranges
Brightness Pattern	Relatively uniform appearance	Higher variation due to lung abnormalities
Image Density	Normal lung structures visible	Presence of dense or opaque regions
Visual Observation	Clearer lung fields	Increased opacity and texture variations
Analysis Outcome	Represents healthy lung patterns	Contains features associated with pneumonia infection

Table 3: Observations from Pixel Intensity Distribution Analysis

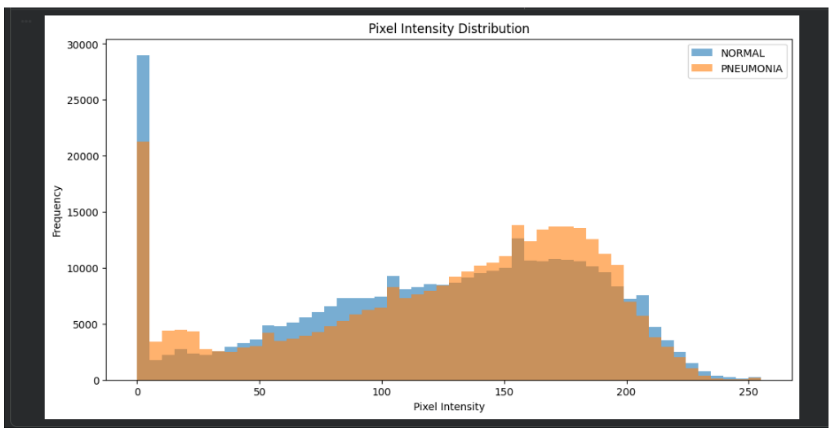

Figure 5: Pixel Intensity Distribution of NORMAL and PNEUMONIA Images

6. Challenges Faced

During the execution of the project, several challenges were encountered while analyzing and understanding the Chest X-Ray dataset.

1. **Dataset Organization and Verification**

   Initially, it was necessary to verify the dataset structure and ensure that all training, testing, and validation folders contained the required image classes. Careful validation was performed to confirm the availability and accessibility of all image files.

2. **Class Imbalance**

   The dataset contained a significantly higher number of PNEUMONIA images compared to NORMAL images. This imbalance was identified during exploratory data analysis and may affect the performance of future machine learning models if not handled appropriately.

3. **Variation in Image Dimensions**

   The chest X-ray images were found to have different resolutions and dimensions. This variability highlighted the need for image standardization and resizing before applying machine learning and deep learning techniques.

4. **Metadata Extraction and Analysis**

   Extracting image metadata such as dimensions, channels, and file formats required processing multiple image samples and organizing the extracted information into a structured format for analysis.

5. **Visualization of Large Datasets**

   Generating meaningful visualizations from a large collection of medical images required careful selection of representative samples and appropriate plotting techniques to ensure clear interpretation of results.

6. **Dashboard Integration**

   Integrating analytical outputs into a React-based dashboard required converting raw analysis results into a format suitable for visualization and presentation. Ensuring consistency between the analysis results and dashboard components was an important part of the development process.

7. Learning Outcomes

- Developed an understanding of the Chest X-Ray Pneumonia dataset and its structure.
- Learned techniques for dataset verification and integrity validation.

- Gained practical experience in extracting and analyzing image metadata.
- Performed Exploratory Data Analysis (EDA) using Python libraries such as Pandas, NumPy, and Matplotlib.
- Learned to visualize and interpret class distribution, image resolution, and pixel intensity patterns.
- Identified dataset challenges such as class imbalance and variation in image dimensions.
- Improved proficiency in handling medical image datasets using Python.
- Gained hands-on experience in creating visualizations and analytical reports.
- Learned the fundamentals of React-based dashboard development for presenting analytical results.
- Established a strong foundation for future machine learning and deep learning tasks related to pneumonia detection.

8. Conclusion

The first week of the internship focused on understanding and analyzing the Chest X-Ray dataset used for pneumonia detection. Dataset verification, integrity validation, metadata extraction, sample image visualization, class distribution analysis, resolution analysis, and pixel intensity analysis were successfully completed. An initial React-based dashboard prototype was also developed to present analytical results in a structured manner. The outcomes of this phase provided important insights into the dataset and created a solid foundation for future stages involving machine learning, deep learning, and automated pneumonia detection.

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